FEKAIRI ELHASSANE

Software Engineer · Go · PostgreSQL · React / React Native

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Summary

Software engineer and fourth-year computer science engineering student at ESI Algiers. Builds Go and PostgreSQL services that stay correct under concurrency and failure (row-level locking, idempotent APIs, transactional outboxes, multi-tenant auth), tested against real databases. Also ships React and React Native clients and has fine-tuned Transformer models for French NLP.

Technical Skills

Languages	Go, TypeScript, JavaScript, SQL, Python, C, C++
Backend	Go (chi, pgx, Bun ORM), Node.js, Express, FastAPI, Flask · REST APIs, concurrency control, idempotency, transactional outbox, job queues, HMAC webhooks, JWT & RBAC, multi-tenancy
Data & Storage	PostgreSQL, MySQL, SQLite, MongoDB, Supabase, Firebase, MinIO (S3)
Frontend & Mobile	React, Next.js, React Native, Expo, TanStack Query, Tailwind CSS, i18next
Testing & DevOps	Testcontainers, Go race detector, pytest, Jest · Docker, Docker Compose, GitHub Actions, Prometheus, golangci-lint, Git
AI / ML	PyTorch, Hugging Face Transformers (CamemBERT, RoBERTa), scikit-learn, pandas, TensorFlow

Experience

Software Developer Intern — DG ONE (EURL) *Sep 2026 · 1 month*

- Built the admin push-notification campaigns feature for DZDash.
- Worked on the wholesaler dashboard and the web version of the client app.

Education

École nationale Supérieure d'Informatique (ESI), Algiers — Computer Science Engineering, Information Systems & Technologies track (currently 2CS, 4th of 5 years) *2023 – 2028 (expected)*

Projects

Transacta – Multi-Tenant Double-Entry Ledger API · github.com/elhassanefek/transacta *Go, PostgreSQL, Docker, Prometheus*

- Atomic multi-leg double-entry transfers and two-phase (pending → posted/failed) transactions; sorted `SELECT … FOR UPDATE` locking prevents lost updates and deadlocks, verified by concurrent integration tests on real PostgreSQL (Testcontainers).
- Idempotency layer on every write: atomic upsert claims, a processing lease for crash recovery, and fencing tokens that discard stale writes; byte-exact replay of cached responses.
- Transactional outbox and reusable Postgres job queue (`SKIP LOCKED`) delivering HMAC-SHA256-signed webhooks with exponential backoff, jitter and dead-lettering; audit trail committed in the same transaction as each ledger change.
- Refresh-token rotation with reuse detection, per-tenant RBAC and isolation; GitHub Actions CI, 95 tests (29 integration) and 4 Architecture Decision Records documenting the trade-offs.

NounouDZ – Childcare Marketplace for Algeria (MVP) *Go, PostgreSQL, React Native, Expo, MinIO*

- Parents and babysitters negotiate immutable, versioned agreements; dual confirmation runs as one row-locked PostgreSQL transaction that creates priced sessions and updates four aggregates in a fixed lock order.
- Made double-booking impossible at the database level with a GiST exclusion constraint on time ranges, proven by concurrent goroutine tests; 263 backend test functions run against containerized PostgreSQL.
- Go worker with a transactional outbox (retries, dead-lettering, stale-lock recovery) and idempotency-key middleware; trilingual (FR/EN/AR) React Native app; 5-job GitHub Actions CI.

Student Feedback Classification (CamemBERT) – LMCS Lab, ESI (co-developed with Samer Kamouche) *Python, PyTorch, Hugging Face*

- Fine-tuned `camembert-base` on 3,056 French student comments (4-label motivation, 3-class opinion); loss weighting and a validation-tuned threshold raised Motivation Macro-F1 from 0.620 to 0.764 (vs. 0.700 for TF-IDF + SVC) and reached 0.807 on Opinion.
- Leak-free pipeline (duplicate detection, grouped stratified splits), fair baseline comparisons, YAML-tracked experiments, 110 pytest tests and Kaggle GPU training.

GDG Events Platform – Admin & Organizer Dashboards *React, TypeScript, shadcn/ui*

- Built the admin and organizer dashboards of the platform used to run GDG events end to end, plus the meal-management module (per-event meals, real-time check-in and attendance tracking).

Other Projects

- **Project Management API** – REST API with role-based JWT auth and real-time task events *(Node.js, Express, MongoDB, Socket.IO)* · GitHub
- **Calm 15** – Interactive CPU emulator for learning computer architecture *(React, GSAP)* · Live
- **Multi-Role E-Commerce Platform** – React frontend with guest, customer and seller/admin views *(React, React Query)* · GitHub